General introduction



Introduction

Typical applications for ACO slot and box channel stainless steel systems include:

- Food processing factories
- Brewing, bottling and canning plants
- Chemical and pharmaceutical industries
- Commercial kitchens
- Schools and leisure
- Retail
- Human and animal healthcare

ACO slot and box channel stainless steel systems are used in applications anywhere where hygienic, corrosion resistance and durable performance are essential.

Why stainless steel?

Stainless steel has the following unique advantages:

- Highly corrosion resistant
- Non-porous, easy to clean and disinfect
- Aesthetically pleasant
- Resistant to temperature extremes and thermal shock
- Coefficient of linear expansion similar to concrete
- 100% recyclable material

All ACO slot and box channels are manufactured from austenitic stainless steel grades 1.4301 (304) or 1.4404 (316L) to EN 10088.

EN 10088	AISI
1.4301	304
1.4404	316L

Table 1

Why ACO slot and box channels?

As one of the leaders of manufacturing stainless steel drainage systems ACO has the knowledge, experience and expertise to offer the optimal drainage solutions for every application. Therefore ACO offers the designer the freedom to choose from a standard range of products as well as bespoke solutions to suit any application.

Surface treatment of stainless steel

The process of cutting, forming and welding of stainless steel will introduce impurities into the surface of the material and unless the appropriate action is taken, the material will begin to corrode and ultimately fail in service. Therefore after fabrication into the final product, it is vital that the stainless steel material is treated with the correct surface treatment to ensure the material is fully corrosion resistant. By applying pickle passivation as the primary surface treatment, the corrosion resistance of stainless steel can be fully restored to its original state, ensuring long and reliable life performance together with the required aesthetic appearance.

Finishes used by ACO include:

- Pickle passivation (acid treatment)
  All ACO channels are pickle passivated by immersing them in a series of acid baths. This is a fundamental requirement for removing iron embedded particulates introduced in the fabrication process and also restoring the chromium depleted regions generated by the welding process. ACO has one of the largest and modern pickle passivation installations in Europe which ensures the optimum corrosion resistance of our products.
- Electropolishing (electrochemical process)
  After pickle passivation, some products are then immersed in an electrolytic fluid in which the products become the anode of a direct current electrical circuit. This process is characterized by the selective attack on the surface of the components whereby upstanding roughness is preferentially dissolved and will yield a progressively smoother, brighter surface.
- Linishing (mechanical process)
  ACO channels have a linished upper edge for esthetical reasons.

Hygienic standards

In order to maintain a clean and sustainable hygienic environment within the food processing area, is therefore essential that drainage products shuld be designed and manufactured to rigorous standards to ensure bacteria traps are minimised.

Slip resistance

The consequences of slips in food processing and kitchen areas can be very serious as accidents can easily occur involving high temperature foodstuffs and cooking surfaces. For applications where the potential of slipping is increased, enhanced slip resistant gratings should be considered. For this reason ACO slot and box channel systems portfolio includes range of slip resistant gratings.

Certification

ACO slot and box channels are fully tested, manufactured and certified in accordance with EN 1433 – Drainage channels for vehicular and pedestrian areas for enhanced operational confidence.

CE

The main and most important benefits of the standard are:

- Load bearing capacity – ensures the installation can safely accommodate vehicle and pedestrian loading conditions in use without damage.
- Water – tightness – management of wastewater in a safe manner without leakage to the surrounding environment. This is particularly important to prevent environmental contamination.
- Durabillity – material selection and design features for a long, trouble-free life.

3rd party product assessment with CE marking to ensure that the products fulfil the Essential Requirements of EN 1433.

Load Class Definition to EN 1433:

Load Class Definition

To EN 1433 – Drainage channels for vehicular and pedestrian areas



Pedestrian and cyclaways, domestic drivesareas.



Car side drainage parking areas, service stations (car), slow-moving light comercial vehicles.



Pedestrian precincts, light vehicles, private car parks and drives.

Range of compatible gullies is certified according to EN 1253 – Gullies in buildings.